A-A-2331
October 1, 1985

COMMERCIAL ITEM DESCRIPTION

INK, DRAWING (WATERPROOF, COLORED)

The General Services Administration has authorized the use of this commercial item description in perference to Federal Specification TT-I-531.

This description covers colored, waterproof drawing ink for general drafting use (drawing and tracing).

The ink shall be of the following types, as specified.

Type I - Transparent
Type II - Opaque

Colors of ink furnished shall be as listed in Table I, as specified.

Salient characteristics.

Composition. Type I, transparent ink, shall be a transparent solution of a dye in a liquid medium. Type II, opaque ink, shall be a suspension of an opaque pigment in a liquid medium, or the ink shall be a combination of transparent dye in solution and opaque pigment in suspension in a liquid medium. The vehicle of the inks shall be an aqueous solution of the necessary ingredients to prevent the precipitation of the dye or the separation of the pigment and to leave a waterproof film on drawing paper. The coloring matter shall be dyes or pigment of suitable color and brilliance, with permanence to light.

Transparency and opacity of ink. Draw lines 1/8-inch wide with black, waterproof drawing ink on white drawing paper. With a brush, paint strips 1/2-inch wide with eacn color over the black lines. If the black lines show clearly through the colors, the inks shall be classified as Type I, transparent dye inks. If the colors substantially obliterate the black lines, the inks shall be classified as Type II, opaque pigments inks.

Stability of ink. The ink shall not show any coagulation, granulation, or precipitation under working or storage condtions. Such evidence of instability shall not be present when the ink is received or after one year storage under normal storage conditions.

Original stability. Examine the sample of ink in its original container for precipitation by inserting a glass rod. If precipitation is present, remove glass rod, replace cap, invert bottle, and shake vigorously for one minute. Examine again for sediment and coagulation. Place a few drops of ink between glass plates, rub plates together and examine for granulation.

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Colors (formulas). Inks of all colors (table I) shall be equal in every essential to inks prepared according to the following formula:

To prepare 100 milliliters of ink, dissolve a quantity of dye as indicated in table I for the respective colors, in 50 milliliters of warm water and add 50 milliliters of shellac solution (100 grams of wax-free, bleached, white shellac per liter of water). Cool to room temperature and filter.

TABLE I. Drawing ink color control formulas*

COLORS	COLOR INDEX	GRAMS OF DYE
	TRANSPARENT INKS	
Blue	C.I. 44095 C.I. Acid blue 97	0.5000
Brown	C.I..30045 C.I. Direct brown 1	0.8000
Carmine	C.I. 18070 C.I. Acid red 40 C.I. 27290 C.I. Acid red 73	0.6400 0.1600
Green	C.I. 42100 C.I. Acid green 9	1.2000
Indigo	C.I. 42735 C.I. Acid blue 104 C.I. 20470 C.I. Acid black 1 C.I. 45380 C.I. Acid red 87	0.7350 0.4350 0.0300
Orange	C.I. 16020 C.I. Acid orange 17	0.6000
Red	C.I. 45425 C.I. Acid red 95	0.5000

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TABLE I, Cont.

Red brick	C.I. 27290	
	C.I. Acid red 73	0.3500
	C.I. 13065	
	C.I. Acid yellow 36	0.1200
	C.I. 42640	
	C.I. Acid violet 49	0.0060
Red violet	C.I. 45440	
	C.I. Acid red 94	0.6487
	C.I. 42640	
	C.I. Acid violet 49	0.1012
Russet	C.I. 21010	
	C.I. Basic brown 4	0.5000
	C.I. 47005	
	C.I. Acid yellow 3	0.3000
	C.I. 16255	
	C.I. Acid red 18	0.2000
Turquoise (Blue-green)	C.I. 42735	
	C.I. Acid blue 104	0.4500
	C.I. 42080	
	C.I. Acid blue 7	0.2000
	C.I. 61570	
	C.I. Acid green 25	0.3500
Vermillion (Red-orange)	C.I. 27290	
	C.I. Acid red 73	0.3080
	C.I. Acid yellow 36	0.0420
Violet	C.I. 42535	
	C.I. Basic violet 1	0.5000
Yellow	C.I. 19555	
	C I. Direct yellow 28	0.4000
Yellow green (Leaf-green)	C.I. 19140	
	C.I. Acid yellow 28	0.6250
	C.I. 42080	
	C.I. Acid blue	0.3750
	OPAQUE INKS*	
White*	—	--

* The ink formulas listed (for transparent inks) will produce aqueous dyestuff solutions considered sufficiently brilliant and intense to be suitable standards for the applicable opaque inks except white. White is the only opaque color for which no comparable transparent ink is possible. White drawing inks require color surveillance by visual examination methods. A method which has been found to be practicable is ruling of several lines of 0.003-inch and 0.020-inch width on a high-contrast blueprint or black bristol board.

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Workmanship. Inks shall be smooth homogeneous liquids free from lumps, strings, and objectionable odor. Labels shall be neatly affixed and clearly printed Ink containers shall not be malformed, chipped, or cracked, and shall show no evidence of spillage or leakage of ink. Ink or containers shall have no defects that affect serviceability or appearance.

Regulatory requirements. In accordance with the section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier's employees to undue hazards from the rewcovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order (except for special marking).

Special marking. Each shipping container shall be marked "KEEP FROM FREEZING" on each side in 1-inch letters. Shipping containers containing glass bottles shall bear "FRAGILE" markings. All shipping containers shall bear arrows and the word "UP" on at least one side and one end and the words "TOP" and "THIS SIDE UP" on the top surface.

Notes. The following terminology included in this commerical item description is defined as follows:

Coagulation - The result of gelation.
Granulation - Hard particles which can be detected by placing a few milliliters of ink between glass plates and then rubbing the plates together.
Precipitation - Precipitates are defined as particles of solids which fall out of solution or suspension and which cannot be redispersed or suspended by either inverting or shaking bottle by hand for 1 minute.

The color index numbers referenced in Table I are listed in the color index available from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.

Purchasers shall specify type and color of ink and size of container.

When this commerical item description is used for procurement, the commercial item certification clause must appear in the solicitation.

PREPARING ACTIVITY:

GSA - FSS

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